

Section 8.2 Exercises

Directions: For exercises 1 - 7, solve each triangle for the unknown side. Round answers to one decimal place.

1. $C = 41.2^\circ, a = 2.49, b = 3.13$

2. $A = 120^\circ, b = 6, c = 7$

3. $B = 58.7^\circ, a = 10.6, c = 15.7$

4. $C = 115^\circ, a = 18, b = 23$

5. $A = 119^\circ, a = 26, b = 14$

6. $C = 113^\circ, b = 10, c = 32$

7. $A = 36.6^\circ, a = 186.2, b = 242.2$

Directions: For exercises 8 - 12, solve for the missing angle of the oblique triangle. Round answers to one decimal place.

8. $a = 42, b = 19, c = 30$; find angle A .

9. $a = 14, b = 13, c = 20$; find angle C .

10. $a = 16, b = 31, c = 20$; find angle B .

11. $a = 13, b = 22, c = 28$; find angle A .

12. $a = 108, b = 132, c = 160$; find angle C .

Directions: For exercises 13 - 18, solve the triangle. Round all answers to one decimal place.

13. $A = 35^\circ, b = 8, c = 11$

14. $B = 88^\circ, a = 4.4, c = 5.2$

15. $C = 121^\circ, a = 21, b = 37$

16. $a = 13, b = 11, c = 15$

17. $a = 3.1, b = 3.5, c = 5$

18. $a = 51, b = 25, c = 29$

Directions: For exercises 19 - 23, use Heron's formula to find the area of the triangle. Round the answer to two decimal places.

19. Find the area of a triangle with sides of length 18 in, 21 in, and 32 in.

20. Find the area of a triangle with sides of length 20 cm, 26 cm, and 37 cm.

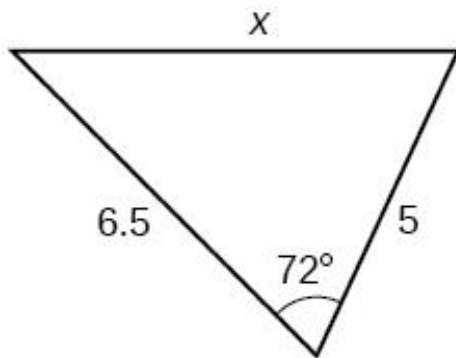
21. $a = \frac{1}{2}$ m, $b = \frac{1}{3}$ m, $c = \frac{1}{4}$ m

22. $a = 12.4$ ft, $b = 13.7$ ft, $c = 20.2$ ft

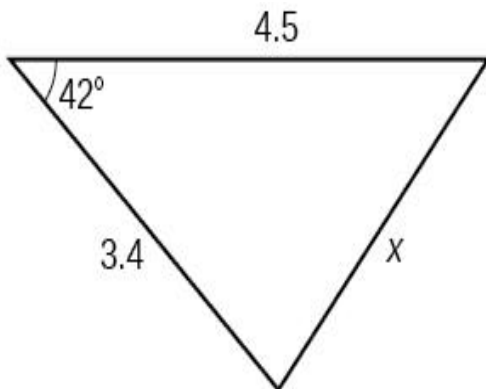
23. $a = 1.6$ yd, $b = 2.6$ yd, $c = 4.1$ yd

Directions: For exercises 24 - 29, find the length of side x . Round the answer to one decimal place.

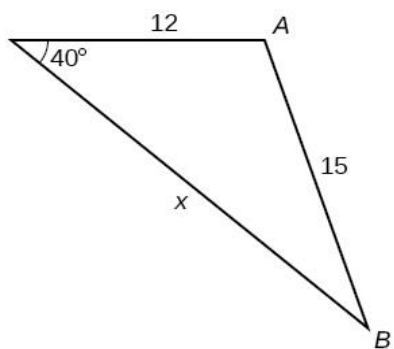
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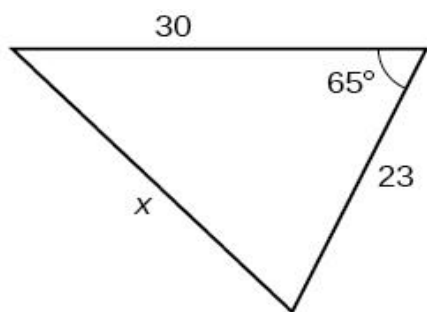
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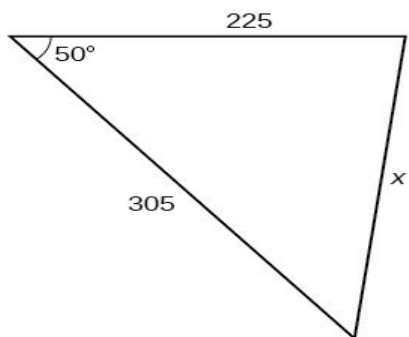
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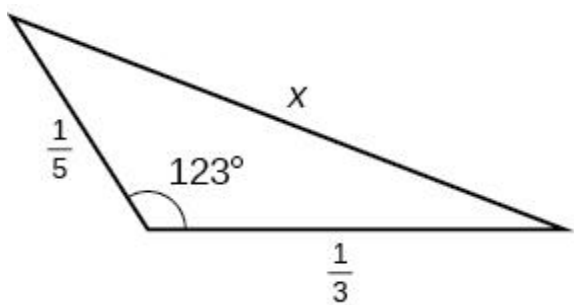
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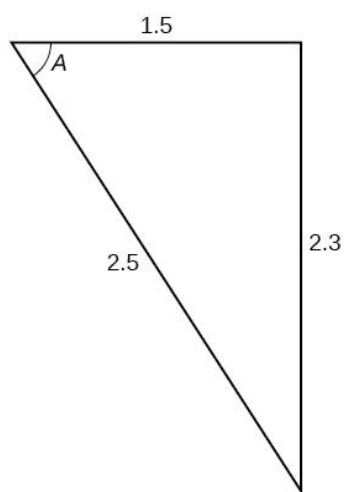


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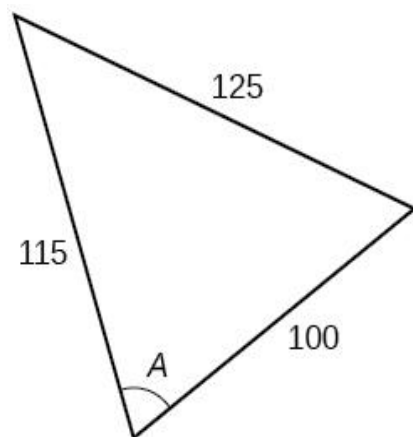


Directions: For exercises 30 - 33, find the measurement of angle A . Round the answer to one decimal place.

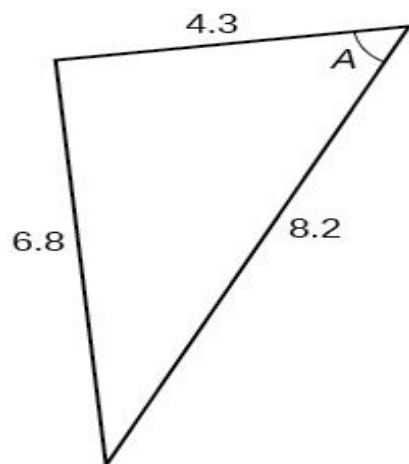
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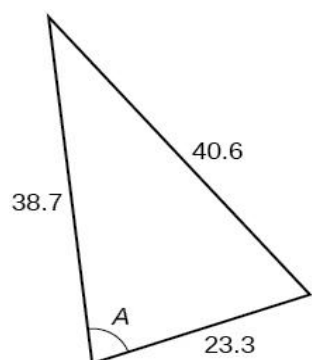
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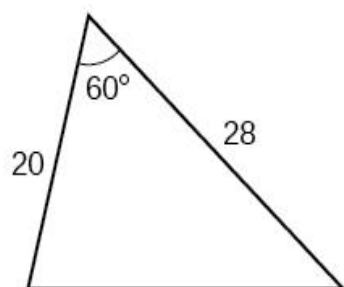


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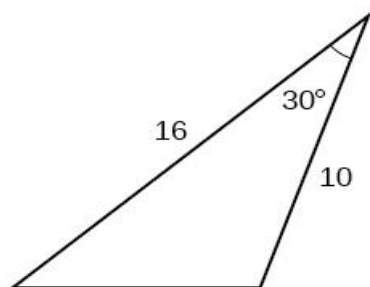


Directions: For exercises 34 - 37, solve for the unknown side. Round the answer to one decimal place.

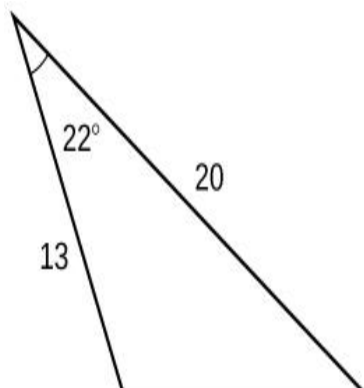
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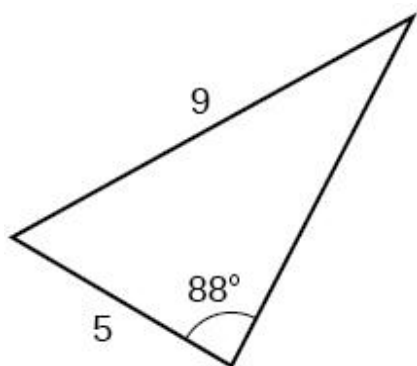
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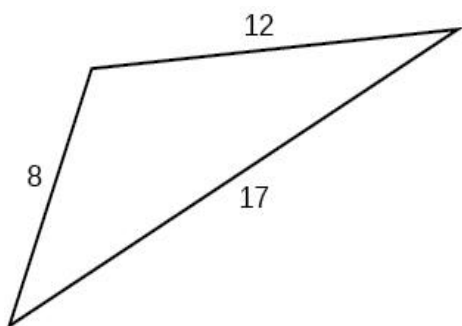


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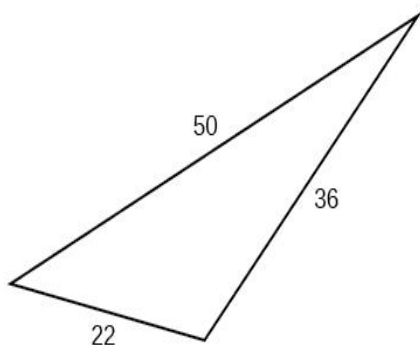


Directions: For exercises 38 - 42, find the area of the triangle. Round the answer to two decimal places.

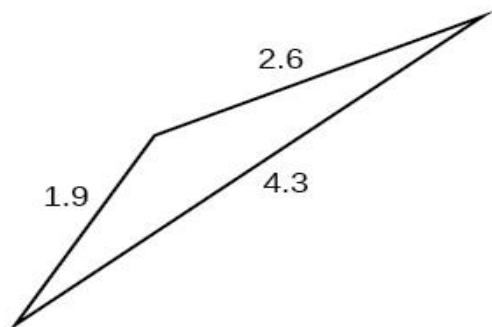
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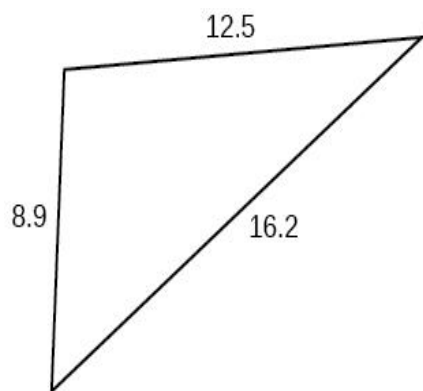
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